

PRACTICALWEEK3

TASK1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
keerthi_priya@LAPTOP-OSGV x + v
keerthi_priya@LAPTOP-OSGVCE7A:~$ mkdir practical3
mkdir: practical3: File exists
keerthi_priya@LAPTOP-OSGVCE7A:~$ cd practical3
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ nano process.c
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ gcc process.c -o process
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ ./process

Parent Process
PID = 634

PPID = 359
Child Process
PID = 635
PPID = 634
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ |
```

```
GNU nano 8.7.1 process
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Process creation failed.\n");
    }

    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());

        sleep(5);
    }

    else
    {
        printf("\nParent Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());

        wait(NULL);
    }

    return 0;
}
```

TASK2: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```

keerthi_priya@LAPTOP-OSGV X + v
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ nano process.c
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ gcc process.c -o process
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ ./process

```

```

Parent Process
PID = 678

```

```

PPID = 359
Child Process
PID = 679
PPID = 678

```

```

keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ ps -el
 F S      UID      PID      PPID    C  PRI   NI     ADDR  SZ  WCHAN    TTY          TIME CMD
 4 S       0        1         0  0   80    0     6079   -   -       ?           00:00:01 system
 5 S       0        2         1  0   80    0     795   -   hvcd0    00:00:00 init-s
 0 S       0        6         2  0   80    0     795   -   hvcd0    00:00:00 init
 4 S       0       49         1  0   79   -1    12599   -   ?        00:00:00 system
 4 S     989       80         1  0   80    0     5604   -   ?        00:00:00 system
 4 S       0       90         1  0   80    0     8861   -   ?        00:00:00 system
 4 S       0      110         1  0   80    0     722   -   ?        00:00:00 chrony
 4 S       0      111         1  0   80    0    1118   -   ?        00:00:00 cron
 4 S     995      112         1  0   80    0    2199   -   ?        00:00:00 dbus-d
 4 S       0      118         1  0   80    0   11024   -   ?        00:00:00 networ
 4 S       0      153         1  0   80    0     4610   -   ?        00:00:00 system
 4 S       0      174         1  0   80    0    448011 -   ?        00:00:00 wsl-pr
 4 S     100      199         1  0   80    0    55137   -   ?        00:00:00 rsyslo
 4 S       0      233         1  0   80    0     1373   -   tty1     00:00:00 agetty
 4 S       0      242         1  0   80    0    30865   -   ?        00:00:00 unatte
 4 S     988      256      110  0   80    0     5954   -   ?        00:00:00 chrony
 1 S     988      264      256  0   80    0     3024   -   ?        00:00:00 chrony
 5 S       0      357         2  0   80    0     796   -   ?        00:00:00 Sessio
 5 S       0      358      357  0   80    0     800   -   ?        00:00:00 Relay(
 4 S    1000      359      358  0   80    0     1567 do_wai pts/0    00:00:00 bash
 4 S       0      360         2  0   80    0     2162   -   ?        00:00:00 login
 4 S    1000      409         1  0   80    0     5628   -   ?        00:00:00 system
 5 S    1000      411      409  0   80    0     5730   -   ?        00:00:00 (sd-pa
 4 S    1000      441      360  0   80    0     1534 poll_s pts/1    00:00:00 bash
 0 R    1000      689      359  0   80    0     1798   -   pts/0    00:00:00 ps

```

```

keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ top
top - 06:01:39 up 8 min, 1 user, load average: 0.15, 0.07, 0.03
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie

```

```
keerthi_priya@LAPTOP-OSGV × + v
4 S 1000 409 1 0 80 0 - 5628 - ? 00:00:00 system
5 S 1000 411 409 0 80 0 - 5730 - ? 00:00:00 (sd-pa
4 S 1000 441 360 0 80 0 - 1534 poll_s pts/1 00:00:00 bash
0 R 1000 689 359 0 80 0 - 1798 - pts/0 00:00:00 ps
keerthi_priya@LAPTOP-OSGVCE7A:~/practical3$ top
top - 06:01:51 up 8 min, 1 user, load average: 0.12, 0.07, 0.03
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 7651.2 total, 6270.5 free, 606.1 used, 927.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7045.1 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24316 15408 11596 S   0.0   0.2   0:01.54 systemd
    2 root        20   0    3180  2204  2072 S   0.0   0.0   0:00.02 init-systemd(Ub
    6 root        20   0    3180  2040  1940 S   0.0   0.0   0:00.00 init
   49 root        19  -1   50396 16732 15448 S   0.0   0.2   0:00.35 systemd-journal
   80 systemd+    20   0   22416 14456 11956 S   0.0   0.2   0:00.13 systemd-resolve
   90 root        20   0   35444 12456  9288 S   0.0   0.2   0:00.71 systemd-udev
  110 root        20   0    2888  1948  1820 S   0.0   0.0   0:00.10 chronyd-starter
  111 root        20   0    4472  3172  2924 S   0.0   0.0   0:00.02 cron
  112 message+   20   0    8796  5416  4728 S   0.0   0.1   0:00.11 dbus-daemon
  118 root        20   0   44096 29700 14556 S   0.0   0.4   0:00.70 networkd-dispat
  153 root        20   0   18440  9396  8148 S   0.0   0.1   0:00.15 systemd-logind
  174 root        20   0 1792044 15264 12616 S   0.0   0.2   0:00.24 wsl-pro-service
  199 syslog      20   0  220548  5860  4564 S   0.0   0.1   0:00.13 rsyslogd
  233 root        20   0    5492  2752  2564 S   0.0   0.0   0:00.02 agetty
  242 root        20   0  123460 32668 18056 S   0.0   0.4   0:00.55 unattended-upgr
  256 _chrony      20   0   23816 11368  7244 S   0.0   0.1   0:00.10 chronyd
  264 _chrony      20   0   12096  2448  1804 S   0.0   0.0   0:00.01 chronyd
  357 root        20   0    3184  1108   980 S   0.0   0.0   0:00.00 SessionLeader
  358 root        20   0    3200  1248  1108 S   0.0   0.0   0:00.05 Relay(359)
  359 keerthi+   20   0    6268  5532  3864 S   0.0   0.1   0:00.09 bash
  360 root        20   0    8648  5556  4712 S   0.0   0.1   0:00.01 login
  409 keerthi+   20   0   22512 13304 10860 S   0.0   0.2   0:00.16 systemd
  411 keerthi+   20   0   22920  4092  2088 S   0.0   0.1   0:00.00 (sd-pam)
  441 keerthi+   20   0    6136  5492  3928 S   0.0   0.1   0:00.03 bash
  701 keerthi+   20   0    8172  6044  3936 R   0.0   0.1   0:00.01 top
```